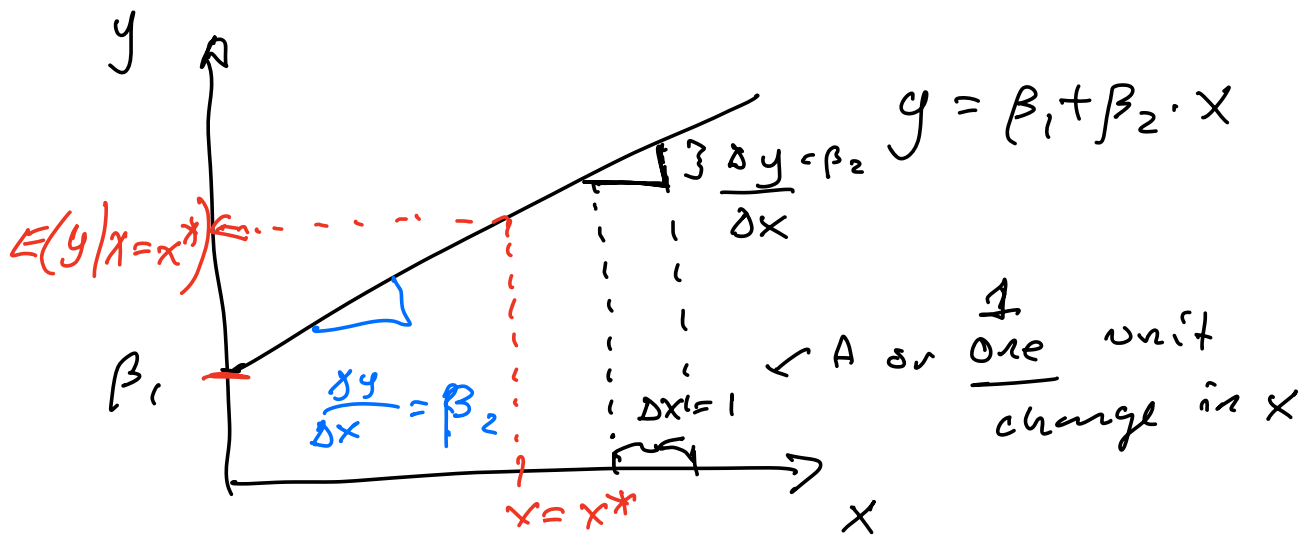




β_1 = value where the fn. crosses the y -axis, when $x = 0$

$$E(y|x=0) = \beta_1 + \beta_2 \cdot 0 = \beta_1$$

$$E(y|x=x^*) = \beta_1 + \beta_2 \cdot x^*$$

constant

$$y = \beta_1 + \beta_2 \cdot x$$

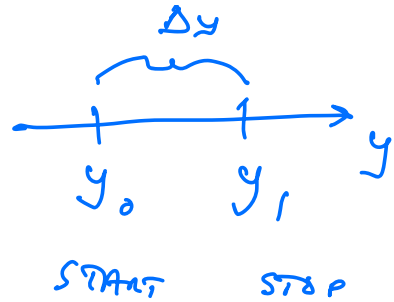
$$dy = d\beta_1 + \beta_2 \cdot dx \quad | \text{ tot diff.}$$

0

$$dy = \beta_2 \cdot dx$$

SLPPG: $\frac{dy}{dx} = \beta_2$ $\frac{dy}{dx} \equiv \frac{\delta y}{\delta x} ?$

$$\text{Elasticity} = \frac{\left(\frac{\Delta y}{y} \right)}{\left(\frac{\Delta x}{x} \right)}$$



$$\% \Delta y = \frac{\Delta y}{y_0}$$

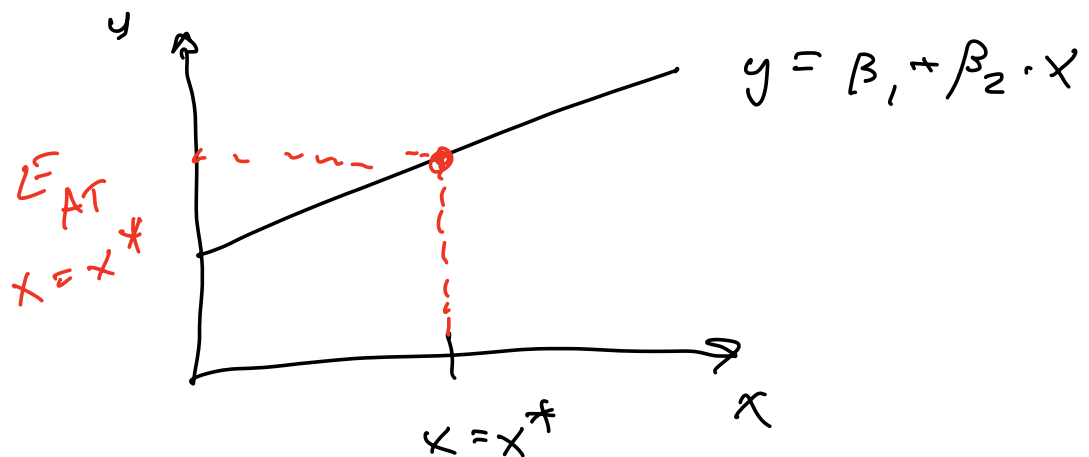
$$= \frac{\Delta y}{y} \cdot \frac{x}{\Delta x}$$

$$y = \beta_1 + \beta_2 \cdot x$$

$$= \underbrace{\frac{\Delta y}{\Delta x}}_{\text{slope}} \cdot \frac{x}{y}$$

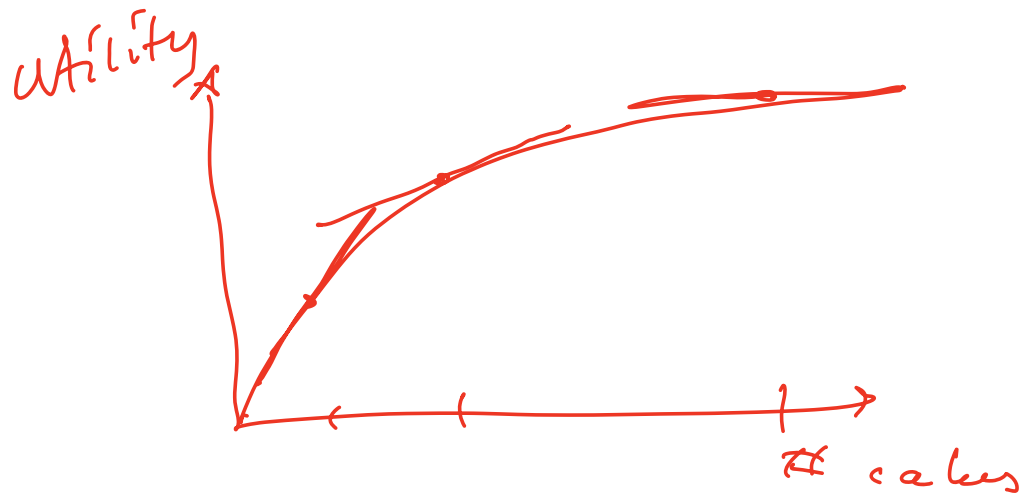
$$\frac{\Delta y}{\Delta x} = \beta_2$$

$$E = \beta_2 \cdot \frac{x}{y}$$



$$E_{AT \ x^*} = \beta_2 \cdot \frac{x^*}{\beta_1 + \beta_2 \cdot x^*}$$

DIMINISHING MARGINAL RETURNS



units →

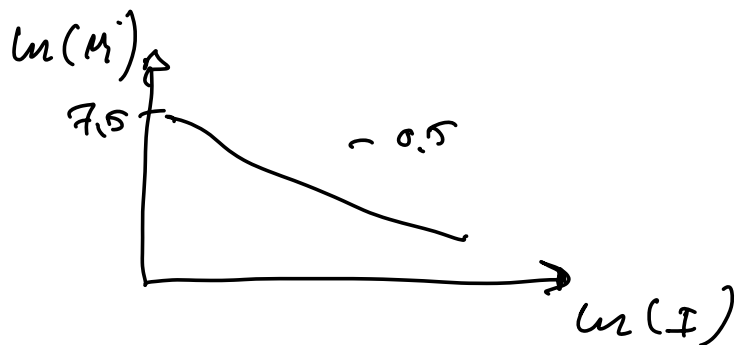
logarithmic transformation

BACK TO LEVELS
USE $M \times \exp(\ln(M))$

M	$\ln(M)$	I	$\ln(I)$
2	0.69	6	
6	1.79	14	
0	∞	0	\$ ∞

mortality

$$\ln(M) = 7.5 - 0.5 \cdot \ln(I)$$



$$d \ln(M) = d(7.5) - 0.5 \cdot d \ln(I)$$

$$\frac{1}{M} dM = 0 - 0.5 \cdot \frac{1}{I} \cdot dI$$

$$\frac{1}{M} dM = -0.5 \cdot \frac{1}{I} dI \quad \left| \cdot \frac{I}{dI} \right.$$

$$\frac{dM}{dI} \cdot \frac{I}{M} = -0.5 \quad (\text{ELASTICITY})$$

FIND THE SLOPE? $\frac{dM}{dI}$

$$\frac{dM}{dI} = -0.5 \cdot \frac{M}{I} \quad \left| \cdot \frac{M}{I} \text{ on both sides} \right.$$

Quadratic m?

$$M = 1400 - 100I + 1.67I^2$$

$$dM = d1400 - 100 \cdot dI + 2 \cdot 1.67 \cdot I \cdot dI$$

$$\frac{dM}{dI} = -100 + 2 \cdot 1.67 \cdot I$$

$$\frac{dM}{dI} \cdot \frac{I}{M} = (-100 + 2 \cdot 1.67 \cdot I) \cdot \frac{I}{M}$$